

Harshana Lakshara Fernando

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github.com/Uchihalthachi | huggingface.co/HarshanaLF | replit.com/@lakshara21
Portfolio: harshanalf.netlify.app

Skills

Programming Languages: C/C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: ReactJS, NextJs, NodeJs, Angular, Rasa, ThreeJs, Tailwind CSS, Lucia, Prisma, NumPy

Databases: PostgreSQL, MySQL, Supabase, Firebase

Technologies & Tools: Jupyter, VS Code, Figma, Hugging Face, OpenAI, Netlify, Vercel, Google Cloud, Nginx, HEDKORU, Docker, docker-compose, GIT, Linux

Testing: Pytest

Automation: GitHub Actions, Makefile, Shell, Batch Scripts

Programming Challenges: [HackerRank Profile](#), [LeetCode Profile](#)

Languages: English, Sinhala

Education

Bsc.(HONS) Engineering(Computer Science and Engineering)

2021 - Present

University of Moratuwa

Relevant Coursework: Object Oriented Programming, Databases, Statistics, Data Structures and Algorithms, Operating Systems, Computer Networks, Programming Languages, Introduction to AI, IOT Devices, Software Engineering, Image Processing, Computer Architecture

High School

2006 - 2019

St John's College, Panadura

Project Work

University Projects

- **Rpal Interpreter** : ("Advanced Compiler Construction and Interpretation")
Demonstrated deep understanding of OOP and data structures and algorithms. Expertise in core CS concepts, including finite automata for lexical analysis (custom regex), pushdown automata for parsing, lambda calculus, and graph theory, particularly in constructing and manipulating ASTs and STs. Proficient in Pytest for testing, Makefile for automation, and exceptional debugging skills, all applied within a robust project framework.
(Tools Used: Python, Visual Studio Code, Command Line, Cygwin, Pytest, GitHub Actions, Makefile)
- **Hotel Assistant Chatbot(Ongoing)** : ("Real-Time AI Conversational Agent")
Developed a real-time conversational AI using the Rasa framework, integrating advanced NLP models and custom components to handle hotel-related queries efficiently. Automated processes with Makefile, ensuring flexibility and extensibility in system design.
(Tools Used: Python, Rasa, Docker, Docker Compose, Nginx, Makefile, Shell Scripts, Batch Scripts)
- **Virtual Hotel Assistant with 3D Animation(Ongoing)** : ("AI-Powered 3D Virtual Assistant")
Implemented a sophisticated virtual assistant that merges 3D animations and AI-driven interactions. The assistant processes user inputs, generates dynamic audio responses, and synchronizes facial animations in real time. Leveraging ThreeJS for 3D animation, Rasa for NLP, and Eleven Labs for TTS, the project showcases cutting-edge visual and conversational AI integration.
(Tools Used: Next.js, Rasa, Eleven Labs(tts), FFmpeg, Rhubarb, Three.js)
- **Opal-System(Ongoing)** : ("Integrated Hotel Management System")
Opal-System is a hotel management system designed for streamlined operations, featuring seamless front-end and back-end development with Next.js. It includes Google OAuth2 for secure user login, uses Supabase for data management, and integrates a Banking API for secure payments, including cryptocurrency support.
(Tools Used: Next.js, Supabase)

Personal Projects

- **H GPT4o(Improving)** : *("Multi-Model AI Assistant Platform")*
H GPT4o is an advanced alternative to ChatGPT 4.0, integrating multiple specialized models and APIs for tasks like image generation, web search, and Q&A. Using the "Duct Tape Method," it combines these components into a flexible, powerful platform. Unlike ChatGPT 4.0, it offers ultimate access but does not have continuous learning.
(Tools Used: *Python, Gradio, Transformers, Hugging Face, PIL, Requests, BeautifulSoup4, Mistral-7B, Meta-Llama-3-8B, Nous-Hermes-2-Mixtral, LLaVA*)
 - **H GO** : *("AI-Powered Chat Interface Inspired by Google Go")*
A sleek chat interface inspired by Google Go, using Hugging Face models and Gradio for real-time responses and easy model switching.
(Tools Used: *Python, Gradio, Nous Hermes Mixtral 8x7B DPO, StarChat2 15b, Mistral 7B v0.3, Phi 3 mini, Mixtral 8x7B*)
 - **Image Captioning and Emotion Analysis** : *("AI-Powered Image Captioning and Sentiment Analysis Tool")*
Gradio app combining Salesforce BLIP for image captioning and RoBERTa for emotion analysis, visualized with bar charts
(Tools Used: *Python, Gradio, Transformers, Matplotlib, PIL, Salesforce BLIP for image captioning, Roberta for emotion classification*)
 - **TubeSummarizer** : *("YouTube Transcript Summarization Tool")*
Gradio app summarizing YouTube video transcripts using YouTube Transcript API, PyTube, and models like Falconsai and Bart
(Tools Used: *Python, Gradio, Transformers, YouTube Transcript API, PyTube*)
 - **Social Media App(Ongoing)** : *("Full-Stack Social Media Platform")*
Full-stack platform inspired by Twitter/Facebook with features like infinite scroll, messaging, likes, and comments.
(Tools Used: *Next.js 15, Shadcn UI, Tailwind CSS, TanStack React Query, Lucia with Prisma adapter, PostgreSQL with Prisma ORM, UploadThing, Stream Chat, Zod, Tip Tap, Vercel*)
 - **3D Portfolio Project(Improving)** : *("Interactive 3D Portfolio with AI Integration")*
Showcases frontend skills with interactive 3D worlds, parallax effects, vertical timelines, and animations using Three.js. Features a 3D avatar with Google Cloud TTS and Mixtral, and includes dark mode.
(Tools Used: *Vite, ReactJs, Tailwind CSS, Three.js, Frame Animation, Mixtral, Google TTS*)
- For more projects and details, visit my profiles on GitHub, Hugging Face, and Replit.

Certificates

- [Python Essential](#)

Future Interests

Generative AI: Advancing skills in text, image, and audio generation.

DevOps: Improving CI/CD and infrastructure automation.

Automation: Optimizing workflows through scripting and tools.

Machine Learning Ops (MLOps): Integrating machine learning models into production environments

Cloud Computing: Expanding cloud-native app development expertise.

REFEREES

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